2.12. Vetores e Momento Relativístico

2.1. Fator de Lorentz p/ $v = 0,6$

$$\gamma = 1,25$$

4-velocidade U^μ

$$U^\mu = \frac{dx^\mu}{d\tau}, \quad d\tau = \frac{dt}{\gamma}$$

$$U^\mu = \gamma \frac{d}{dt}(t, \vec{x}) = \gamma (1, \vec{v})$$

Logo: $U^\mu = (1,25;\ 0,75;\ 0;\ 0)$.

4-momento p^μ

$$p^\mu = mU^\mu = m(1,25;\ 0,75,\ 0,\ 0)$$

$$p^\mu = (1,25\,m;\ 0,75\,m,\ 0,\ 0)$$

$\underline{U^\mu U_\mu = -1}$

$$U^\mu U_\mu = \eta_{\mu\nu} U^\mu U^\nu = -(U^0)^2 + (U^1)^2 + (U^2)^2 + (U^3)^2$$

$$= -(1,25)^2 + (0,75)^2 + 0^2 + 0^2 = -1$$

Logo: $U^\mu U_\mu = -1$ //

$\underline{p^\mu p_\mu = -m^2}$

$$p^\mu p_\mu = \eta_{\mu\nu} p^\mu p^\nu = -(p^0)^2 + (p^1)^2 + (p^2)^2 + (p^3)^2 = -m^2.$$ //

$\underline{K = E - m}$, $\quad E = p^0 = \gamma m = 1,25m$ //

a) Pion π $\begin{cases} \text{massa} = m_\pi \\ \text{4-mom} = p_\pi'^\mu \end{cases}$ (em S')

Fótons γ $\begin{cases} \text{massa} = m_\gamma = 0 \\ \text{4-mom} = p_1'^\mu \text{ e } p_2'^\mu \end{cases}$

Para o pion $p_\pi'^\mu = (m_\pi, 0, 0, 0)$

Para os fótons: $E' = |\vec{p}'| \rightarrow p'^\mu = (E', \underbrace{\vec{p}'}_{\text{3-mom}})$, e

$$p_1'^\mu = (E_1', p_1'^x, 0, 0) \quad \text{e} \quad p_2'^\mu = (E_2', p_2'^x, 0, 0)$$

Conservação do 4-mom:

$$p_\pi'^\mu = p_1'^\mu + p_2'^\mu$$

* Parte espacial:

$$0 = \vec{p}_1' + \vec{p}_2' \rightarrow \vec{p}_1' = -\vec{p}_2'$$

Implicando em $|\vec{p}_1'| = |\vec{p}_2'| \rightarrow E_1' = E_2' = E_\gamma'$

* Parte temporal:

$$m_\pi = E_1' + E_2' = 2E_\gamma' \rightarrow \boxed{E_\gamma' = \frac{m_\pi}{2}}$$

b) Em S(laboratório)

$\gamma = 10 \rightarrow \gamma \approx 0{,}994987$

$p_1'^\mu = (m_\pi/2, m_\pi/2, 0, 0) \longrightarrow +x$

$p_2'^\mu = (m_\pi/2, -m_\pi/2, 0, 0) \longleftarrow -x$

Calculando a energia de cada fóton em S:

$$E = \gamma(E' + v p'^x)$$

$$E_1 = \gamma(E_1' + v p_1'^x) = \gamma\left(\frac{m_\pi}{2} + v\frac{m_\pi}{2}\right) = \gamma\frac{m_\pi}{2}(1+v)$$

$$E_1 = \frac{10 \cdot m_\pi}{2}(1 + 0{,}994987)$$

$$E_1 \approx 9{,}975\ m_\pi$$

$$E_2 = \gamma(E_2' + v p_2'^x) = \gamma\frac{m_\pi}{2}(1 - v) = 5\,m_\pi(1 - 0{,}994937)$$

$$E_2 \approx 0{,}025\ m_\pi$$

Assim:

Fóton pra $+x$ $\longrightarrow$ $\boxed{E_1 \approx 9{,}975\ m_\pi}$

Fóton pra $-x$ $\longrightarrow$ $\boxed{E_2 \approx 0{,}025\ m_\pi}$

c) $$P^0 = E_1 + E_2 = 9{,}974937\,m_\pi + 0{,}025063\,m_\pi$$
$$= 10{,}00000\ m_\pi.$$

$$P^1 = E_1 - E_2 = 9{,}974937\,m_\pi - 0{,}02563\,m_\pi.$$
$$= 9{,}549874\,m_\pi$$

$$M^2_{sist} = -P^\mu P_\mu = (P^0)^2 - |P|^2 = (10{,}00000\,m_\pi)^2 + - (9{,}949874\,m_\pi)^2$$

$$M^2_{sist} = (100 - 99{,}00000)\,m_\pi^2 = 1{,}00000$$

Logo $M_{sist} = m_\pi$. (Numéricamente)

2.3. ① $\gamma \longrightarrow \quad \longleftarrow e^-$

② $\gamma \longleftarrow \quad \longrightarrow e^-$??

① Fóton inicial $p_\gamma^\mu = (E, E, 0, 0)$

Elétron inicial $p_e^\mu = (E_e, -p_e, 0, 0)$

$p_e = \sqrt{E_e^2 - m_e^2}$

② Fóton recuado (180°) $p_\gamma'^\mu = (E', -E', 0, 0)$

Elétron recuado (?) $p_e'^\mu = (E'_e, p'_e)$

Pela conservação do 4-momento

$\bar{p}_e' + \bar{p}_\gamma' = \bar{p}_e + \bar{p}_\gamma$ $\quad \bar{p}$ são vetores

$\bar{p}_e' = \bar{p}_e + \bar{p}_\gamma - \bar{p}'_\gamma$

$\bar{p}_e'^2 = (\bar{p}_e + \bar{p}_\gamma - \bar{p}_\gamma')^2$, com $p_e^2 = p_e'^2 = -m_e^2$ e $p_\gamma^2 = p_\gamma'^2 = 0$

$-m_e^2 = (\bar{p}_e^2 + 2\bar{p}_\gamma \cdot \bar{p}_e + \bar{p}_\gamma^2 - 2(\bar{p}_e + \bar{p}_\gamma)\cdot \bar{p}_\gamma' + \bar{p}_\gamma'^2)$

$-m_e^2 = p_e^2 + 2\bar{p}_\gamma \bar{p}_e + \cancel{p_\gamma^2}_0 - 2\bar{p}_e \bar{p}_\gamma' - 2\bar{p}_\gamma \bar{p}_\gamma' + \cancel{p_\gamma'^2}_0$

$\cancel{-m_e^2} = \cancel{-m_e^2} + 2\bar{p}_\gamma \cdot \bar{p}_e - 2\bar{p}_e \bar{p}_\gamma' - 2\bar{p}_\gamma \cdot \bar{p}_\gamma'$

$0 = 2(\bar{p}_e \cdot \bar{p}_\gamma - \bar{p}_e \cdot \bar{p}_\gamma' - \bar{p}_\gamma \cdot \bar{p}_\gamma')$

Assim: $\bar{p}_e \cdot \bar{p}_\gamma' + \bar{p}_\gamma \cdot \bar{p}_\gamma' = \bar{p}_e \bar{p}_\gamma$.

* $p_e \cdot p_\gamma = -E_e E + (-p_e)(E) = -E(E_e + p_e)$.

* $p_\gamma \cdot p_\gamma' = -EE' + E(-E') = -2EE'$

* $p_e \cdot p_\gamma' = -E_e E' + (-p_e)(-E') = -E'(E_e - p_e)$.

Então:

$$-E'(E_e - p_e) - 2EE' = -E(E_e + p_e)$$

$$E'((E_e - p_e) + 2E) = E(E_e + p_e)$$

$$\boxed{E' = E\left(\frac{E_e + p_e}{E_e - p_e + 2E}\right)}$$

No limite ultravelativístico: $E_e \gg m_e \gg E$, que nos leva a $\gamma_e = E_e/m_e \gg 1$. Assim:

$$p_e = E_e \cdot v_e$$

Substituindo em E'.

$$E' = E\left(\frac{E_e(1 + v_e)}{E_e(1 - v_e) + 2E}\right) \quad : E_e$$

$$E' = E\left(\frac{1 + v_e}{(1 - v_e) + 2E/E_e}\right)$$

Em Taylor. $v_e = \sqrt{1 - \frac{1}{\gamma_e^2}} \approx 1 - \frac{1}{2\gamma_e^2}$.

Assim, podemos considerar:

- $1 + v_e \approx 2$
- $1 - v_e \approx 1/2\gamma_e^2$.

Então: $E' \approx E\left(\frac{2}{1/2\gamma_e^2 + 2E/E_e}\right) \approx E\left(\frac{2}{(1/2\gamma_e^2 + 2E/(\gamma_e m_e)}\right)$

Multiplicando por $2\gamma_e^2$

$$\boxed{E' = \frac{4\gamma_e^2 E}{1 + \frac{4\gamma_e E}{m_e}}}$$

Situacão 1: $\gamma_e E << m_e$

$$\frac{4\gamma_e E}{m_e} << 1 \rightarrow 1 + \frac{4\gamma_e E}{m_e} \approx 1.$$

Assim: $E' \approx 4\gamma_e^2 E$

A energia do fóton espalhado cresce com γ_e^2.

Situacão 2: $\gamma_e E >> m_e$.

$$1 + \frac{4\gamma_e E}{m_e} \approx \frac{4\gamma_e E}{m_e}$$

Assim: $E' \approx \frac{4\gamma_e^2 E}{4\gamma_e E/m_e} = \gamma_e m_e = E_e$

A energia do fóton é (praticamente) toda a energia do elétron. O elétron "para" //

2.5. $\frac{M}{M_0} \approx e^{-\phi}$

<u>Foguete (M, P^μ, U^μ)</u> $\qquad P^\mu = M U^\mu$

Em termos de rapidez: $U^\mu = (\cosh\phi, \operatorname{senh}\phi, 0, 0)$

$$P^\mu = (M\cosh\phi, M\operatorname{senh}\phi, 0, 0)$$

<u>Fótons</u> $(m_\gamma = 0,\ E = |\bar{p}|, -x)$

$$dP^\mu_\gamma = (dE_{rad}, -dE_{rad}, 0, 0)$$

Conservação do 4-momento

$$P^\mu = (P^\mu + dP^\mu) + dP^\mu_\gamma$$

$$dP^\mu_\gamma = -dP^\mu = -d(MU^\mu) = -(dU^\mu M + U^\mu dM)$$

Como fótons são "tipo-luz", $P_\gamma \cdot P_\gamma = 0$

$$(dP_\gamma)_\mu (dP_\gamma)^\mu = 0 \rightarrow (d(MU^\mu))_\mu (d(MU^\mu))^\mu = 0$$

$$(d(MU^\mu))^2 = (dM)^2(\cancelto{-1}{U_\mu U^\mu}) + 2MdM\cancelto{0}{U_\mu dU^\mu} +$$

$$+ M^2(dU_\mu dU^\mu) = 0$$

$$(d(MU^\mu))^2 = M^2(dU_\mu dU^\mu) = 0$$

Como: $U^\mu = (\cosh\phi, \operatorname{senh}\phi, 0, 0)$, temos:

$$dU^\mu = (\operatorname{senh}\phi, \cosh\phi, 0, 0)$$

$$dU_\mu dU^\mu = -(\operatorname{senh}^2\phi\, d\phi)^2 + (\cosh^2\phi\, d\phi)^2$$

$$= \underbrace{(\cosh^2\phi - \operatorname{senh}^2\phi)}_{1}(d\phi)^2 = (d\phi)^2$$

$$(-dM)^2 + 2MdM(0) + M^2(d\phi)^2 = 0$$

$$(dM)^2 = M^2(d\phi)^2$$

$$dM = \pm M d\phi$$ (Pegando o negativo só, pois $dM<0$)!

$$dM = -M d\phi$$

$$\int_{M_0}^{M} \frac{dM}{M} = -\int_0^{\phi} d\phi$$

$$e^{\left(\ln\left(\frac{M}{M_0}\right)\right)} = e^{-(\phi)}$$

$$\boxed{\frac{M}{M_0} = e^{-\phi}}$$

b) $$v = \tanh\phi = \frac{e^{\phi}-e^{-\phi}}{e^{\phi}+e^{-\phi}}$$

$$v(e^{\phi}+e^{-\phi}) = e^{\phi}-e^{-\phi}$$

$$ve^{\phi} + ve^{-\phi} = e^{\phi} - e^{-\phi} \quad \cdot(e^{\phi})$$

$$ve^{2\phi} + v = e^{2\phi} - 1$$

$$e^{2\phi}(-v+1) = 1 + v$$

$$e^{2\phi} = \frac{1+v}{1-v}$$

$$\left(\frac{1}{e^{-\phi}}\right)^2 = \frac{1+v}{1-v}$$

$$\left(\frac{M_0}{M}\right)^2 = \frac{1+v}{1-v} \rightarrow \boxed{\frac{M}{M_0} = \sqrt{\frac{1-v}{1+v}}}$$

c) $v = 0{,}9$

$$\frac{M}{M_0} = \sqrt{\frac{1-0{,}9}{1+0{,}9}} = \sqrt{\frac{0{,}1}{1{,}9}} \approx 0{,}2294$$

Quando $v = 0{,}9$ $\approx 23\%$ da massa do foguete ainda "está lá". O restante foi convertido em fótons.